

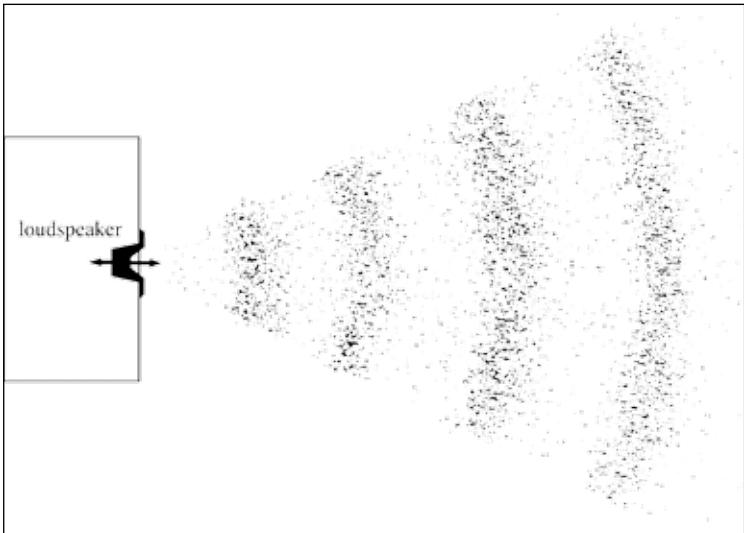
Loudspeakers are one of the most ubiquitous pieces of technology around us. Whether we realise it or not, they've become a huge part of our lives. From that annoying PA system blaring on the road to the drool worthy home theatre surround sound system you lust for, or the very unseen laptop speakers, we rely on loudspeakers without ever understanding just how they work, or how we could make our experience with them even better.

If you're trying to understand how loudspeakers accurately reproduce sound, you'll first have to understand how sound is actually produced.

How is sound produced?

Sound is produced when something vibrates, which transmits the kinetic energy to the air particles around it. A vibrating particle creates vibrations in the medium around it, be it air or water. When this medium is air, these vibrations are known as travelling longitudinal waves, and these are the waves we can hear as sound waves. Sound waves, when analyzed, will reveal to have areas of high pressure, known as compressions, and low pressure, known as rarefactions.

Frequency is the word we use to describe the speed of a single wave. The frequency is affected by the speed of vibration of the sound-creating object.



Sound waves in air